

Sonic Sensemaking: Augmenting Narrative Integration of Invisible Intellectual Processes

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Figure 1: 24echoes: a design probe that enables revisiting the flow of a day, including time periods that are difficult to capture in task records, by presenting short ambient-audio snippets segmented by hour. Each cell represents one hour; selecting a cell plays the corresponding audio.

ABSTRACT

The purpose of this study is to facilitate sensemaking—the integration of subjective experiences into a “self-narrative”—by using fragmented environmental sounds to reclaim moments often lost to efficiency-driven values. In modern intellectual labor, daily value is often dominated by objective results, causing invisible cognitive processes and subtle social interactions to be forgotten as “worthless” due to a lack of tangible output. While AI-driven lifelogs attempt to verbalize experiences, they often impose fixed meanings and hinder reflection on information that is difficult to process as language. We propose a method to induce a process where users actively retrieve memories and reconstruct meaning by intentionally providing ambiguous auditory fragments with lowered information resolution.

We developed a system that presents 30-second environmental sound fragments via web-based interfaces and propose a 3-day auto-ethnography to investigate how these interactions evolve over time. By analyzing the retrieval frequency of non-task memories and their impact on subjective value, we aim to clarify the process of constructing self-narratives independent of productivity metrics. The success criteria include the rediscovery of subjective meaning in previously discarded time and the qualitative enrichment of daily journaling.

Index Terms: Personal informatics, reflection, lifelogging, ambient audio, design probe.

1 INTRODUCTION

In contemporary society, the value of an individual’s day is predominantly measured by objective and quantitative metrics, such as the

volume of completed tasks and tangible outputs achieved. Especially within the realm of intellectual labor, efficiency has become the primary yardstick for self-evaluation. However, this output-centric perspective often fails to capture significant aspects of daily life, such as prolonged periods of thought that did not reach a conclusion, the subtle influence of the work environment on cognition, or casual social interactions. These “unproductive moments” constitute the core of an individual’s unique cognitive habits and internal processes. Yet, because they are difficult to quantify, they are frequently forgotten, leading to a hollow self-assessment of “having done nothing” despite a day filled with diverse experiences.

To address this challenge, the Human-Computer Interaction (HCI) community has proposed various visual lifelogging systems to support reflection. For instance, PhotoClock [2] and Chronoscope [3] synchronize past photographs with temporal information to facilitate the recall of visual memories. However, photographs are specialized in recording “events” curated by the photographer’s intent. Consequently, they cannot capture the “voids” where no photos were taken or the “internal cognitive states” that are inherently invisible. More recently, approaches using Large Language Models (LLMs) to automatically summarize lifelogs and verbalize the meaning of a day have gained popularity. While these tools reduce the cognitive load of reflection, they risk causing “Narrative Closure”—the premature fixation of an experience’s meaning into a single interpretation. As Baumer et al. [1] point out, the core of reflection lies in the user maintaining “Reflective Agency.” Delegating the act of meaning-making to an AI may deprive users of the opportunity to polysemically reinterpret their experiences and discover new personal significance.

In this research, we move away from visual information and linguistic summarization to focus on an approach that leverages the inherent ambiguity of sound. Petrelli et al. [8] demonstrated in their work on “sonic mementos” that sound can be more evocative than photographs, sparking imagination and richer conversations. Similarly, Odom et al.’s “Olly” [7] utilizes music listening histories to prompt reminiscence. However, these existing auditory interventions primarily target “special occasions” (e.g., vacations) or

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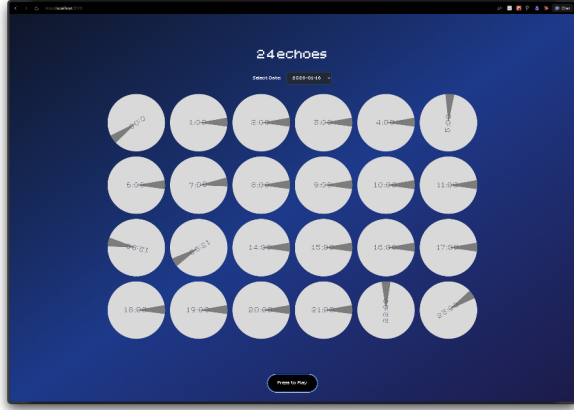


Figure 2: Main window of 24echoes. The 24 hours of a day are represented as 24 discs arranged in a 4x6 grid, where each disc links to a short ambient-audio snippet from the corresponding hour.

finished content (e.g., music) for emotional reminiscence. They do not address the daily intellectual production process or the sense-making of the invisible, mundane moments that fill our days.

The objective of this research is to utilize the “low information resolution” and “contextual richness” of daily environmental sounds to design a “cognitive whitespace” where users can actively construct meaning by referencing their own memories. As Gaver et al. [4] propose, “Ambiguity” can serve as a vital resource for design by increasing user engagement in interpretation. Environmental sounds demand an active, “puzzle-solving” cognitive effort, where meaning is generated through the interaction between the sound and the user’s memory. Specifically, we developed “24echoes,” a Design Probe that presents short environmental sound fragments sampled from each hour of the day, and evaluated it using an Autobiographical Design [6] approach. In this paper, we describe and analyze how the author’s recall and daily narratives evolved through long-term interaction with the system. Through this, we clarify how multi-layered self-understanding independent of productivity metrics can be formed and propose design guidelines for interfaces that preserve “room for re-interpretation.”

2 SYSTEM DESIGN

This section describes the design details of “24echoes,” a Design Probe intended to support the sensemaking of invisible daily cognitive processes through non-verbal environmental sounds. The system reconfigures a single day into 24 discrete auditory cues to elicit active interpretation from the user.

2.1 Visual Interface and Temporal Structure

The core of the interface is a 4x6 spatial grid that maps the 24-hour temporal structure (Figure 2). Each cell contains a rotating disk-shaped object, corresponding to the hours from 0:00 to 23:00, arranged from top-left to bottom-right.

The system extracts a random 30-second audio segment for each hour from continuous environmental sound data recorded via wearable devices. If no data exists for a specific hour, the corresponding cell remains visible, but the audio is processed as a 30-second silent segment. Furthermore, the background color of the interface transitions subtly according to the time of day (e.g., amber for morning, cyan for midday, indigo for night), serving to complement the user’s Time Awareness alongside the abstract auditory information.

2.2 Interaction Design and Reflective Agency

A primary design priority for 24echoes was maintaining the user’s “Reflective Agency.” According to the Reflective Agency Framework proposed by Baumer et al. [1], three elements are essential for technology to appropriately support user reflection:

- **Internal Origination:** The motivation or initiation of reflection should arise from the user, not the system.
- **Calibrated Responsiveness:** The system should respond to the user’s state without imposing excessive meaning or intervention.
- **Reflective Ambiguity:** The system should avoid providing definitive answers and instead allow for polysemic interpretations by the user.

Based on this framework, the system does not utilize “push” notifications to force the listening of environmental sounds. While the presence of a record is always visually indicated by the rotation of the disks (Calibrated Responsiveness), the choice of whether to actually listen to the sound is left entirely to the user (Internal Origination). Additionally, the system does not verbalize or summarize the audio; by presenting the environmental sounds as a medium that leaves room for interpretation, it encourages meaning-making linked to the user’s own memories (Reflective Ambiguity).

2.3 Ambient Feedback and Amplitude Analysis

To support “exploration” before a user commits to listening, the system implements ambient feedback that analyzes the real-time amplitude of the audio and converts it into the rotation speed of the corresponding disk. In quiet environments (e.g., reading, late-night work), the disk rotates slowly, whereas in noisy environments (e.g., commuting, conversation), it rotates rapidly.

This visual feedback continues even when the audio is muted. This allows the user to peripherally perceive the “dynamic nature of that time” through visual movement before actually listening. By utilizing “Ambiguity as a Resource” as proposed by Gaver et al. [6], information is not presented as a symbolic “summary” but as a Resource for Investigation, prompting users to actively explore moments that pique their interest.

2.4 Implementation and Architecture

The system infrastructure leverages Amazon Web Services (AWS) to ensure secure and scalable data management. Audio data recorded by the wearable device is uploaded to an S3 bucket via strict access control using Presigned URLs, ensuring that the back-end APIs and storage are accessible solely by the authenticated user.

3 USER STUDY AND DISCUSSION

We positioned 24echoes as a Design Probe and conducted a three-day autoethnography with the first author as the participant (Figure 3). Throughout the study period, we collected continuous 24-hour audio recordings and confirmed that 30-second audio segments for each hour were ingested into the system without missing data. In this section, we report three observations: (1) recollection of activities not captured by To-Do lists or calendars, (2) the relationship between the type of information contained in audio and what is recalled, and (3) how the 30-second segment length and the listening timing influence reflection. We then summarize design implications based on these observations.

3.1 Recalling Activities Not Represented in To-Do Lists or Calendars

When listening to the ambient audio for each hour with 24echoes, there were moments in which the first author recalled activities that are not explicitly represented in To-Do lists or calendars (e.g.,

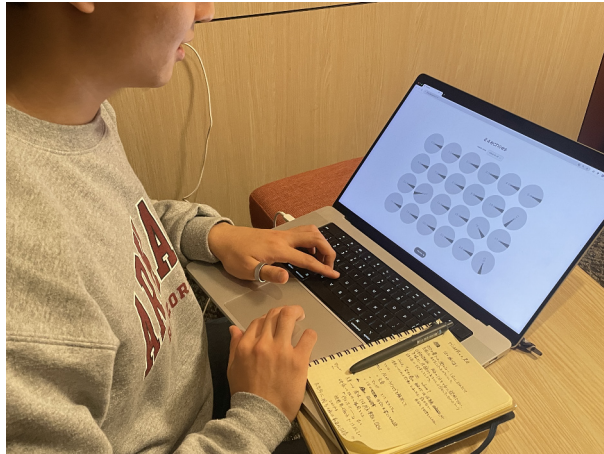


Figure 3: Study setup. A participant interacted with 24echoes on a laptop, selecting hourly cells to play short ambient-audio snippets and taking notes on recalled events and context during the reflection.

waiting during transit, preparing for work, or browsing information while background music was playing). Although these activities did not directly correspond to tangible outputs, they were articulated as factors that explained time allocation and how work progressed throughout the day. In addition, even when listening to recordings from several days earlier, differences across hourly soundscapes served as cues to identify “what I was doing” and “where I was,” and the day’s events were described as a continuous flow. This observation suggests that time that is difficult to account for solely in terms of outcomes or task completion (e.g., preparation, waiting, transit, and intermediate steps) can become a target of reflection when supported by short audio segments.

3.2 Relationship Between Information Type in Audio and What Is Recalled

What was recalled varied depending on the type of information contained in the ambient audio (the first author’s speech, others’ conversations, or environmental noise).

- **Conversation (audio containing linguistic information):** When the first author’s own speech was present, the topic being handled at the time, the rationale for decisions, and the affective state immediately prior were described in conjunction with the spoken content.
- **Environmental noise (primarily non-linguistic audio):** Sounds such as platform crowd noise or train departure chimes primarily served as cues to identify place, movement, and purpose (i.e., why the author was there), rather than eliciting detailed thoughts.

Furthermore, we observed that even audio segments containing only noise or silence functioned as effective retrieval cues when presented with their temporal context. For instance, a segment recording “white noise” at 14:00 triggered the specific memory of “leaving the device in the lab to attend a lecture in another building.” Similarly, silence late at night revealed “falling asleep with the TV on.” These inferences suggest that in the short term (e.g., within 24 hours), the combination of precise timestamps and raw audio—even without semantic content—allows users to deductively reconstruct the “voids” of their day, a process that might be lost if an algorithm automatically discarded “low-quality” audio.

Overall, these observations indicate that audio containing linguistic information may contribute to recovering “content” (what one was thinking or discussing), while primarily non-linguistic audio may contribute to identifying “context” (where one was and for what purpose).

3.3 Effects of 30-Second Segment Length and Listening Timing

At the same time, the design choice to segment each hour into 30 seconds imposed constraints on how reflection unfolded. For example, in situations where the author wished to trace complex feelings or a line of reasoning in detail, the switch to the next hour after 30 seconds tended to shift attention toward checking subsequent time slots rather than deepening interpretation within the current segment. We also observed differences by listening timing. When listening to recordings from the same day at the end of the day, the author more readily mapped fragments of audio to recent events and added descriptions that bridged connections across hours. Taken together, these observations suggest that a 30-second window is better suited to comparing hours and selecting which moments to attend to, rather than reconstructing a single episode in fine detail.

3.4 Ethical Considerations

We acknowledge the privacy concerns associated with continuous audio recording. For data collection, we utilized the “Limitless” pendant [5], which features a visible LED indicator during recording to notify surroundings and a manual stop function to ensure user control. While this study was an autoethnography with full author consent, we recognize that future deployments involving broader participants must include strict protocols for bystander consent, such as geofencing to disable recording in private spaces or utilizing on-device voice masking before cloud upload.

3.5 Limitations and Future Work

As this study positions “24echoes” as a design probe to explore the design space of sonic reflection, the evaluation was limited to a short-term autoethnography (N=1). We do not claim these findings generalize to a broader population or prove the system’s quantitative efficacy compared to existing tools. Future work will involve comparative studies with larger sample sizes, contrasting this ambient approach against traditional text diaries and visual lifelogs. Specifically, we aim to investigate how “bystander consent” can be managed in social settings and whether the “random sampling” strategy remains robust over longer durations (e.g., weeks or months) where detailed memory fades.

4 CONCLUSION

In this work, we designed and implemented 24echoes as a design probe that supports reflection without summarizing or fixing a day into language. Instead, the system presents short segments of ambient audio as cues for revisiting one’s day. 24echoes represents a 24-hour day as a 24-cell grid, providing a 30-second audio segment for each hour.

For an initial evaluation, we conducted a three-day autoethnography with the first author, using continuous 24-hour recordings and observing what was recalled while interacting with the system. From these observations, we found that (1) activities that are not typically represented in To-Do lists or calendars (e.g., waiting, preparation, and transit) became targets of recall; (2) what was recalled differed depending on the nature of the audio—segments containing linguistic conversation tended to elicit recalled content related to thoughts or feelings, whereas primarily non-linguistic environmental sounds tended to function as cues for identifying context such as place and purpose; and (3) the 30-second time window and the timing of listening influenced how reflection unfolded, with 30-second segments being used more readily for comparing hours

and forming an overview of the day than for tracing a single episode in fine detail.

At the same time, these findings are based on an initial exploration with a single participant (the first author) over a short period (three days), which limits generalizability. Future work should examine longer-term use with participants in diverse situations and more systematically investigate how parameters such as segment duration and listening interactions shape recall and reflection.

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REFERENCES

- [1] E. P. S. Baumer, J. P. Khovanskaya, M. Adams, J. P. Pollak, T. Volda, and G. Gay. Reviewing reflection: On the use of reflection in interactive system design. In *Proc. DIS '14*, pp. 93–102, 2014. 1, 2
- [2] A. J. S. Chen, D. S. Y. Cheng, and W. Odom. Photoclock: Reliving memories in digital photos as the clock ticks in the present moment. In *Proc. DIS '23*, pp. 1957–1971, 2023. 1
- [3] A. J. S. Chen, H. Y. S. Sang, and W. Odom. Exploring memory-oriented interactions with digital photos in and across time: A field study of chronoscope. In *Proc. CHI '23*, 2023. 1
- [4] W. W. Gaver, J. Beaver, and S. Benford. Ambiguity as a resource for design. In *Proc. CHI '03*, pp. 233–240, 2003. 2
- [5] Limitless AI. Limitless pendant. <https://www.limitless.ai/>, 2024. Accessed: 2026-01-30. 3
- [6] C. Neustaedter and P. Sengers. Autobiographical design in hci research: Designing and learning through use-it-yourself. In *Proc. DIS '12*, pp. 514–523, 2012. 2
- [7] W. Odom, A. J. S. Chen, H. Y. S. Sang, and P. L. P. Cheng. Investigating slowness as a frame to design longer-term experiences with personal data: A field study of olly. In *Proc. CHI '19*, 2019. 1
- [8] D. Petrelli, N. Villar, E. Kalnikaite, J. Dib, and S. Whittaker. Fm radio: Family interplay with sonic mementos. In *Proc. CHI '10*, pp. 2371–2380, 2010. 1